Matrix functions and stability

A uniform disk bound for wave-kernel Padé approximants

Difficulty: hard **Importance:** interesting to specialist

Status: Lean verified

Rating rationale: Hard because this is a focused all-orders Padé inequality within established approximation theory; its immediate impact is on specialist wave-kernel error analysis.

Resolution — 2026-09-11

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For every integer $m \geq 1$, the normalized diagonal Padé denominator for $\cosh \sqrt{z}$ is nonzero on $|z| \leq 3$ and $|1 - r_m(z)| \leq 2$ there. The bound is strict for $m \geq 2$ and sharp for $m = 1$ at $z = 3$. The analytic tail argument and exact finite certificates cover every order.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below. The later Lean proof and verification evidence records the exact target and kernel checks.

Primary reference: complete authored PDF, standalone TeX, **Theorem 1**. Independent proof review · Authorship and submission record. This manuscript review is an independent agent review, not external human peer review; the later formal checks are recorded separately.

Lean proof and verification evidence

Reviewed 13 October 2026.

At immutable repository revision *bca7904b*, *NLA.Proofs.MF03.all_order_target* has the literal frozen type *NLA.Statements.MF03.Target*. The independent final statement and proof review checks every positive Padé order, existence of a reduced representative, all reduced representatives, denominator nonvanishing on the closed radius-three disk, and the error bound two. Audited finite certificates cover orders 1–15, and the source determinant argument covers every order at least 16; the proof adds no premise or weaker radius.

The manifest pins Lean **4.33.1**, Mathlib *0df444a360ea*, and LeanCert *621a43d7cf21*. From *lean-statements*, run *lake build NLA.Proofs.MF03.AllOrderTarget NLA.Proofs*. The successful PR #349 aggregate kernel build ran the imported target and its *#assert_statement* and *#assert_trust kernel* checks. The independent audit reports only *propext*, *Classical.choice*, and *Quot.sound* as transitive axioms, with no proof admission or custom axiom. This catalog reviewed those dated public and independent records; it did not rerun Lean locally for this status update.

Problem statement

Let

$$f(z) = \sum_{j=0}^{\infty} \frac{z^j}{(2j)!} = \cosh \sqrt{z},$$

where the series defines the entire function without a square-root branch choice. For each integer $m \geq 1$, let $r_m = P_m/Q_m$ be its diagonal Padé approximant at zero: $\deg P_m, \deg Q_m \leq m$, $Q_m(0) = 1$, and $Q_m(z)f(z) - P_m(z) = O(z^{2m+1})$. Is it true that the reduced rational function r_m has no pole in $\{z \in \mathbb{C} : |z| \leq 3\}$ and

$$|1 - r_m(z)| \leq 2 \qquad (|z| \leq 3)$$

for every m?

Reference and status evidence

Nadukandi and Higham, Computing the Wave-Kernel Matrix Functions, SIAM J. Scientific Computing 40(6) (2018), DOI, §4.2, Conjecture 4.6, manuscript p. 12. Lemma 4.5 establishes the finite range $m \leq 20$; §4.3 explains its role in backward-error analysis. Searches for the paper title with "conjecture" and for "Conjecture 4.6" with "cosh" and "Padé" located no general proof or counterexample. The status evidence is therefore weaker than a recent paper explicitly reaffirming the conjecture.

Audit — 2026-09-10

Rechecked Lemma 4.5 and Conjecture 4.6: the displayed assertion is proved for $1 \leq m \leq 20$, while arbitrary m remains conjectural. Paper-title and conjecture-number searches found no general resolution. The open portion still rests on historical primary evidence.